Enterprise V2.0 | Custom AI Engineered by SaNDS Lab | UNV Cameras Recommended (Highly Compatible)

Bahrain & Gulf Labor Law Compliant | SaNDS Lab HR Solution Sync | 100% Offline AI | Multi-Gate RTSP

— SANDS LAB MIDDLE EAST W.L.L • CUSTOM AI ENGINEERED SOLUTIONS

CCTV AI Multi-Gate Face Attendance & Security Surveillance System

Next-generation contactless biometric workforce attendance, live visitor movement intelligence, and active security threat surveillance powered by **Custom AI Engineered by SaNDS Lab**. Fully synchronized with **SaNDS Lab Middle East HR Solution** and architected for 100% compliance with **Bahrain & Gulf (GCC) Labor Law regulations** with automated overtime, Ramadan schedules, and payroll ledger exports.

100% OFFLINE & PRIVATE (ZERO CLOUD)	**< 0.4s** SUB-SECOND RECOGNITION SPEED
Dual-Gate AUTOMATED IN / OUT TRIP LOGIC	**99.4%** CUSTOM AI BIOMETRIC ACCURACY

Custom AI Engineered by SaNDS Lab Middle East W.L.L

Unlike generic cloud APIs or off-the-shelf facial recognition gadgets, this solution features a **proprietary computer vision architecture engineered directly by SaNDS Lab**. Built specifically for complex facility entrances, uneven illumination, natural walking-speed throughput, and continuous CCTV streams, delivering enterprise-grade accuracy with zero cloud latency and complete data sovereignty.

Executive Summary & Commercial Value

Value Proposition

Traditional biometric attendance systems rely on touch-based fingerprint readers or facial terminals installed at doorframes, resulting in morning bottlenecks, hardware wear-and-tear, maintenance costs, and hygiene concerns. Cloud-based facial recognition software introduces recurring monthly SaaS subscriptions, external network latency, and severe data privacy compliance liabilities under GCC Personal Data Protection Laws (PDPL) and GDPR.

SaNDS Lab CCTV AI Multi-Gate Face Attendance solves all these challenges through an innovative, non-intrusive paradigm: workers and visitors walk naturally through entrances while standard security CCTV cameras capture live video streams. The local Python AI engine detects and identifies faces in real-time on-premise, tracks entry/exit transitions across multi-gate pairs, compiles daily timesheets with outing break analytics, alerts security against blacklisted intruders, and syncs seamlessly with corporate HR payroll databases.

RECOMMENDED CAMERA BRAND

UNV Cameras: Highly Compatible & Strongly Recommended

UNV (Uniview) CCTV Cameras are highly compatible and strongly recommended by SaNDS Lab for deployment with our AI face attendance system. UNV cameras deliver outstanding RTSP stream consistency, sub-second packet transit times, high-fidelity True WDR (Wide Dynamic Range), and superior optical clarity across changing ambient light conditions, ensuring peak face recognition accuracy.

Universal Multi-Brand Compatibility: While **UNV Cameras are recommended** for optimum performance and factory-tested stability, the SaNDS Lab system is also **fully compatible with other major CCTV brands** including **Hikvision, Dahua, Axis Communications, Hanwha Vision, and all standard ONVIF (Profile S/G/T) compliant IP Cameras**.

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Core Product Features & Capabilities

V2.0 Feature Matrix (100% On-Premise)

CUSTOM AI ENGINEERED

100% Offline Deep Learning AI

Custom AI Engineered by SaNDS Lab to run fully on-premise without transmitting any image or biometric data over the internet. Features zero recurring cloud API charges, zero external latency, and complete data privacy compliance.

- ✓ **Dual-Stage Architecture:** Custom Engineered By SaNDS Lab AI Engine integrating OpenCV YuNet (~232 KB) detector and SFace (~38 MB) 128-dimensional embedding model.
- ✓ **Adaptive Normalization:** Robust recognition across variable lighting, angles, head tilt, and high-res mobile photo enrollments.
- ✓ **Anti-Duplication Throttle:** Configurable cooldown timer (e.g. 60 seconds) prevents multiple log spamming for standing personnel.

UNV RECOMMENDED

Multi-Camera Gate Pairing (IN / OUT)

Connects to unlimited standard IP CCTV cameras over standard RTSP (Port 554). Supports grouping cameras into logical gate pairs representing physical doors, gates, turnstiles, and emergency exits.

- ✓ **UNV Cameras Recommended:** UNV (Uniview) cameras are highly compatible and strongly recommended for lowest frame jitter and ultra-stable 24/7 RTSP streams.
- ✓ **Other Brands Compatible:** Also 100% compatible with Hikvision, Dahua, Axis, and all ONVIF-compliant CCTV cameras.
- ✓ **Directional Role Assignment:** Cameras designated specifically as Entry (`IN`) or Exit (`OUT`).
- ✓ **Live Stream Watchdog:** Automatic reconnection and ping health monitoring for online/offline status.

AUTOMATED LOGIC

Multi-Trip Timesheet & Outing Analytics

Eliminates the confusion of multiple punches during a working day. The backend calculation engine chronologically aggregates all punches into accurate, enterprise-grade daily timesheets.

- ✓ **First IN & Last OUT:** Accurately tracks official start and departure times for shift compliance.
- ✓ **Outing & Trips Counter:** Automatically detects every mid-day exit (e.g. lunch breaks, meetings, errands).
- ✓ **Productive Hours vs. Breaks:** Computes exact net inside work hours vs. cumulative outside break minutes.
- ✓ **Shift Policy Evaluation:** Automatic flagging of Present, Late, Half-Day, Under-Time, or Absent status.

NEW IN V2.0

Guest & Visitor Movement Tracking

Transform passive CCTV cameras into an intelligent visitor management solution without requiring manual sign-in books or front-desk reception bottlenecks.

- ✓ **Auto Guest Profiling:** Unenrolled faces detected at entry gates automatically receive a unique tracking code (`GST-XXXX`).
- ✓ **Movement Timeline:** Complete audit trail of entry time, gate location, stay duration, and exit punch.
- ✓ **Visitor Categorization:** Tag guests as Clients, Contractors, Maintenance, Delivery, or VIPs with company and phone info.

NEW IN V2.0

Blacklist & Security Breach Alerting

Proactive perimeter security surveillance. Safeguards premises by maintaining a biometric watchlist of banned personnel, disgruntled former employees, and security risks.

- ✓ **Threat Level Classification:** Categorized as Critical, High, Medium, or Low threat with incident rationale.
- ✓ **Instant HUD Breach Alert:** Real-time pop-up modal on operator screens with snapshot photo, gate location, and alarm.
- ✓ **Incident Resolution Workflow:** Security officers can log dispatched guard actions and mark incidents as Resolved.

V1.1 SPRINT

Live Video Monitor & Floating Toast HUD

Comprehensive real-time command center interface designed for security guard kiosks and HR reception screens with rich visual feedback.

- ✓ **Live Face Overlay:** Dynamic bounding boxes, identity tags, and confidence percentages over video stream.
- ✓ **Corner Punch Toast:** Floating toast in the bottom-right corner displaying verified profile photo, name, and IN/OUT badge.
- ✓ **Admin UI Toggle:** Complete administrator control to toggle floating popups on/off in System Settings.

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System Architecture & Data Pipeline

Zero Cloud Topology

1

CCTV RTSP Ingest

UNV Recommended (also Hikvision, Dahua, Axis, ONVIF) stream RTSP H.264/H.265 via Port 554.

2

SaNDS AI Worker

Custom Engineered by SaNDS Lab AI Engine (YuNet detector + SFace 128-d vector matcher).

3

PHP REST API

Validates punch cooldowns, matches IN/OUT gates, logs snapshots, and updates MySQL.

4

React UI & HR Sync

Displays live HUD popups, calculates timesheets, and synchronizes with HR database.

NATIVE BRIDGE

SaNDS Lab Middle East HR Solution 2-Way Sync

Native bi-directional data synchronization with **SaNDS Lab Middle East HR Solution** (`hr_solution_db`). 1-Click import of staff CPR numbers, designations, and biometric profile photos; automated daily export of verified timesheets directly into corporate payroll ledger with audit logs.

LABOR LAW COMPLIANT

Bahrain & Gulf (GCC) Labor Law Engine

Built specifically to comply with **Bahrain Labour Law (Law No. 36 of 2012)** and GCC unified labor regulations. Automates statutory 8h/day (48h/week) caps, daytime overtime (125%), night/weekend overtime (150%), mandatory 6-hour continuous work break enforcement, and LMRA-ready audit trails.

SEASONAL SCHEDULER

Ramadan Reduced Hours & GCC Shifts

Supports Standard Office, School Class, Flexible, Night, and statutory **Ramadan Reduced Hours (6 hours/day, 36 hours/week)** with customizable grace periods for late check-in and early departure without salary deduction penalties.

DATA GRID

Universal Fast DataTable (V1.1)

High-performance grid across all 7 views with instant multi-column search, dynamic sorting, customizable page sizes (10 to 100 rows), and 1-click Excel/CSV export for payroll administrators.

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Technical Specifications & Hardware Compatibility

Deployment Specs

Software Architecture Stack

- **AI Engine:** Custom AI Engineered by SaNDS Lab (Python 3.10+, OpenCV DNN YuNet + SFace ONNX).
- **Frontend Application:** ReactJS 18, Vite Bundler, Vanilla Modern CSS, Lucide Icons.
- **Backend Middleware:** PHP 8.1+ REST API, PDO Secure MySQL Driver, CORS Middleware.
- **Database:** MySQL 8.0+ / MariaDB with UTF8MB4 Full International Unicode support.
- **Streaming Protocol:** RTSP (Real-Time Streaming Protocol) H.264 / H.265 on Port 554.

CCTV Compatibility & Deployment Standards

- **Recommended Cameras:** **UNV (Uniview) IP Cameras are Highly Compatible & Strongly Recommended** for proven sub-second RTSP decoding and 24/7 stream resilience.
- **Universal Brand Support:** Also fully compatible with Hikvision, Dahua, Axis, Hanwha Vision, and all standard ONVIF Profile S/G/T IP Cameras.
- **Dual-Stream Optimization:** Sub-stream (640×360 @ 10–15 FPS) for AI face detection; Main-stream (1080p) for high-res archival snapshots.
- **Camera Positioning:** Mounted at 2.0m – 2.5m height, facing entrance walk-path at a 10°–15° downward angle.

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GCC Localization, Labor Law & Corporate Branding

Enterprise Ready

FEATURE / CAPABILITY	SUPPORTED SPECIFICATIONS	STATUS
SaNDS Lab HR Solution Bridge	Automated 2-way sync with `hr_solution_db`: CPR sync, designation mapping, automated attendance export to payroll ledger	✓ Native 2-Way Sync
Bahrain Labour Law (Law 36/2012)	Statutory 8h/day & 48h/week limits, 125% daytime overtime, 150% night/weekend overtime, 6h continuous break tracking, LMRA audit records	✓ 100% Compliant
Gulf (GCC) Labor Framework	UAE Federal Decree-Law 33/2021, Saudi Labor Law M/51, Qatar, Kuwait, Oman regional labor standards	✓ 100% Compliant
Ramadan Seasonal Hours Engine	1-Click automated shift scheduler reducing official working time to statutory 6 hours/day (36h/week) for fasting employees	✓ Automated
Camera Brand Certification	**UNV (Uniview) Highly Compatible & Strongly Recommended**; other brands (Hikvision, Dahua, Axis, ONVIF) also 100% compatible	✓ UNV Certified
GCC & India Timezones	Asia/Bahrain, Asia/Dubai, Asia/Riyadh, Asia/Qatar, Asia/Kuwait, Asia/Muscat, Asia/Kolkata (IST)	✓ Active
Multi-Currency Formats	BHD (Bahraini Dinar), SAR, AED, QAR, KWD, OMR, INR (₹), USD ($)	✓ Active
Date & Time Formats	Configurable `DD/MM/YYYY`, `YYYY-MM-DD`, `MM/DD/YYYY` | 12-Hour (AM/PM) or 24-Hour Military Time	✓ Active
Corporate Theme Customizer	Custom Header Color, Menu Accent Color (Default Royal Blue `#1e40af` & Pink `#ec4899`), Light/Dark Switcher	✓ Configurable
Custom Brand Assets	Official SaNDS Lab logo integration, custom organization title, and mandated corporate footer disclaimer	✓ Branded

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Product Comparison: Why SaNDS Lab CCTV AI Wins

Market Comparison

Evaluation Criteria	SaNDS Lab CCTV AI (V2.0)	Fingerprint / RFID Terminals	Cloud Facial SaaS
SaNDS Lab HR Solution Sync	Native 2-Way Sync (hr_solution_db)	Manual Excel Export / Import	Costly Custom API Connector
Bahrain & GCC Labor Compliance	100% Built-in (Bahrain Law 36/2012)	Manual Spreadsheet Adjustments	Generic US/EU Rules Only
Camera Brand Compatibility	UNV Recommended (Other Brands Also Compatible)	Proprietary Terminals Only	Restricted Cloud Hardware
Recurring Monthly Software Fees	Zero ($0 Monthly Fees)	Zero ($0)	High Monthly Subscriptions
Biometric Privacy & GDPR / PDPL	100% On-Premise LAN	Local Hardware Storage	High Risk (Cloud Database)
Contactless / Hygiene / Walking Speed	100% Contactless (Natural Walk)	Physical Touch Bottlenecks	Contactless
Existing CCTV Hardware Utilization	Yes (Standard RTSP IP Cameras)	Requires Dedicated Terminals	Requires Dedicated Terminals
Outing Breaks & Multi-Trip Calculation	Automatic IN/OUT Gate Pairing	Manual IN/OUT Button Selection	Basic Single Punch Only
Visitor Tracking & Blacklist Alerts	Built-in (Active HUD Alerts)	Not Supported	Requires Costly Add-ons

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Server Hardware Sizing & Capacity Planning (10 – 50 Cameras)

UNV Cameras Recommended | Validated Benchmark Matrix

Computational throughput scales predictably with concurrent RTSP video decoding channels and facial feature extraction cycles. UNV (Uniview) cameras are strongly recommended for optimal frame pacing and low latency; other brands (Hikvision, Dahua, Axis, ONVIF) are also fully compatible. Below are the verified server configurations for **SaNDS Lab Custom Engineered AI Engine** across 10, 20, 30, 40, and 50 camera tiers:

CAMERA TIER	PROCESSOR (CPU)	RAM	GPU / AI ACCELERATION	STORAGE ARCHITECTURE	NETWORK, CHASSIS & POWER
10 Cameras (5 IN / 5 OUT Gates)	**Intel Core i7-13700** (16C / 24T) or Xeon E-2378 (8C/16T)	**32 GB** DDR4/DDR5 3200MHz	**NVIDIA RTX 3060 (12GB)** or Intel UHD 770 OpenVINO (CPU only)	**512 GB NVMe M.2 SSD (OS & AI)** **+ 2 TB SATA SSD (Logs/Photos)**	Dual 1 Gbps RJ45 LAN Tower / 1U Rack (500W PSU)
20 Cameras (10 IN / 10 OUT Gates)	**Intel Core i9-13900K** (24C / 32T) or Xeon Silver 4314	**64 GB** DDR5 ECC / Non-ECC	**NVIDIA RTX 4070 (12GB)** or NVIDIA RTX A2000 (12GB Workstation)	**1 TB Enterprise NVMe (RAID 1)** **+ 4 TB Enterprise SATA (RAID 1)**	Dual 2.5 Gbps / 10 Gbps SFP+ 1U / 2U Rack (650W PSU)
30 Cameras (15 Gate Pairs)	**Intel Xeon Silver 4416+** (20C / 40T) or AMD EPYC 7302 (16C)	**64 GB** DDR5-4800 Registered ECC	**NVIDIA RTX 4080 (16GB)** or NVIDIA RTX A4000 (16GB PCIe)	**2× 1 TB Enterprise NVMe (RAID 1)** **+ 8 TB Enterprise HDD (RAID 5)**	Dual 10 Gbps SFP+ Optical 2U Rack (Redundant 750W 1+1)
40 Cameras (20 Gate Pairs)	**Intel Xeon Gold 5418Y** (24C / 48T) or AMD EPYC 7443 (24C)	**64 GB – 128 GB** ECC Registered	**1× NVIDIA L4 (24GB)** or 2× NVIDIA RTX 4060 Ti (16GB each)	**2× 1 TB Enterprise NVMe (RAID 1)** **+ 12 TB Storage (RAID 5)**	Dual 10 Gbps SFP+ Fiber 2U Rack (1+1 Redundant 800W)
50 Cameras (25 Gate Pairs)	**Dual Intel Xeon Gold 6330** (56C / 112T) or AMD EPYC 7763 (64C)	**128 GB** DDR4/DDR5 ECC Reg	**2× NVIDIA L4 (24GB x 2 = 48GB)** or 2× NVIDIA RTX 4000 Ada (32GB VRAM)	**2× 2 TB Enterprise NVMe (RAID 1)** **+ 16 TB – 24 TB Storage (RAID 6/10)**	Dual 10 Gbps SFP+ Redundant 2U Enterprise Rack (1+1 1100W)

💡 **High-Density Architectural Best Practices (30 - 50 Cameras):**

- **Camera Recommendation:** UNV (Uniview) cameras are strongly recommended for factory-validated RTSP stream integrity; other brands (Hikvision, Dahua, Axis, ONVIF) are also 100% compatible.
- **Dual-Stream Ingest:** Configure cameras to feed Sub-stream (640×360 @ 10–15 FPS) for AI face detection; grab primary 1080p frame only upon face match for audit logging (reduces decoding load by >60%).
- **Multi-Worker Sharding:** Dedicated Python processes pinned across CPU/GPU cores prevent single camera timeouts from affecting adjacent gate channels.
- **Recommended Platforms:** Dell PowerEdge R650 / R750, HPE ProLiant DL380 Gen10/Gen11, or Supermicro GPU Server A+ series.

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